

Exact stability of affine sections of Hilbert balls and the sharp role of uniform continuity

Candidate full extension for arXiv:1902.02363

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Status: candidate full answer, likely valid. For closed balls in arbitrary Hilbert spaces, affine fibers of a closed-range operator have an explicit orthogonal-product form and an exact Hausdorff distance. This removes compactness and ambient finite dimensionality from the source example. Arbitrary continuous objectives remain valid when the kernel is finite-dimensional; uniformly continuous objectives work for every kernel. An explicit bounded continuous objective shows that mere continuity is not enough when the kernel is infinite-dimensional.

1 Source question

The source is Daniel Reem, Simeon Reich, and Alvaro De Pierro, *Stability of the optimal values under small perturbations of the constraint set*, arXiv:1902.02363v4 [1]. Example 5.11 proves continuity of

$$t \longmapsto \inf\{f(x) : x \in C, Lx = t\}$$

when $C \subset \mathbb{R}^n$ is compact, convex, and has nonempty interior and $f : C \rightarrow \mathbb{R}$ is continuous. In Remark 5.12(i), page 19, the authors write:

“We believe that the theory developed in this section can be extended further, and, in particular, that it is possible to remove (at least in some interesting cases) the compactness and finite-dimensionality assumptions from Example 5.11.”

The result below supplies such a case, with an exact geometric modulus, and also determines precisely where the source proof’s use of compactness cannot simply be discarded.

2 Proof intuition

Let $N = \ker L$ and $M = N^\perp$. Closed range makes the restriction $L|_M$ continuously invertible onto $L(H)$. Thus fixing $Lx = t$ fixes the M -coordinate of $x - c$, while the N -coordinate ranges over a ball. Pythagoras gives both its center and radius. Two fibers are therefore orthogonal translates of concentric kernel-balls, so their Hausdorff distance is the Euclidean combination of center displacement and radius displacement.

This exact formula yields continuity, local Lipschitz bounds, and value stability for uniformly continuous objectives without any compactness. If N is finite-dimensional, the fibers encountered along a convergent parameter sequence lie in one compact set, so mere continuity is again enough. If N is infinite-dimensional, orthogonal directions allow localized continuous wells to escape without accumulating. Those wells make the value function jump even though the fibers converge in Hausdorff distance.

Below we collect a few remarks regarding Examples 5.9 and 5.11.

Remark 5.12. (i) Examples 5.9 and 5.11 extend partly, but significantly, the stability theory developed in [89, pp. 279–280] and [87, Theorem 2] (for a related theory, see [89, p. 281] and [77, Lemma 4.1]). This theory has been applied to analyzing stochastic programs [87, Section 4], [77, Section 4]. The setting in [89, pp. 279–280] and [87, Theorem 2] is a finite-dimensional Euclidean space, a polyhedral set C , and an objective function f which is Lipschitz continuous on a set which contains C ; another requirement in [89, pp. 279–280] is that either the level-sets of f are bounded or $C^\infty \cap A_0 = \{0\}$, where A_0 is the kernel of the linear operator L and C^∞ is the horizontal cone associated with C . It is proved in [89] that the associated optimal value function ϕ_* is Lipschitz continuous under these assumptions (note: the constraint set there is written as $\{x \in C : Lx = b - t\}$, where b is a given vector and t is the parameter; thus by a simple change of variable we can arrive at this formulation). It can be seen that Examples 5.9 and 5.11 extend this theory to the case of spaces which are not necessarily normed spaces (and not necessarily finite-dimensional), a constraint set C which is either the entire space or a (usually non-polyhedral) convex body, a linear operator L which should have a Lipschitz continuous EGI, and an objective function f which is either Lipschitz continuous or merely uniformly continuous. We are still able to derive the continuity of ϕ_* under these conditions, and sometimes (Example 5.9) its Lipschitz continuity.

We believe that the theory developed in this section can be extended further, and, in particular, that it is possible to remove (at least in some interesting cases) the compactness and finite-dimensionality assumptions from Example 5.11.

(ii) The set I mentioned in Example 5.11 is actually open and convex whenever it is nonempty. Indeed, given $s \in I$, let $x \in E_s \cap \text{Int}(C)$. In particular, $x \in \text{Int}(C)$ and

Figure 1: The source question in full-width context, page 19.

3 Exact fiber geometry

Let H, Y be real Hilbert spaces and let $L : H \rightarrow Y$ be bounded and linear with closed range $R = L(H)$. Put

$$N := \ker L, \quad M := N^\perp,$$

and let

$$G : R \longrightarrow M$$

be the inverse of $L|_M : M \rightarrow R$. This inverse is bounded by the bounded inverse theorem [2]. Fix $c \in H$, $r > 0$, and the closed ball

$$C := \overline{B}(c, r).$$

For

$$J := \{t \in R : \|G(t - Lc)\| \leq r\}, \quad I := \{t \in R : \|G(t - Lc)\| < r\},$$

define

$$A_t := C \cap L^{-1}(t), \quad z_t := G(t - Lc), \quad \rho_t := \sqrt{r^2 - \|z_t\|^2} \quad (t \in J).$$

Here I is exactly the set of parameters whose affine fiber meets $\text{Int } C$.

Theorem 1 (Exact affine-section formula). *For every $t \in J$,*

$$A_t = c + z_t + \rho_t \mathbb{B}_N, \quad \mathbb{B}_N := \{n \in N : \|n\| \leq 1\}. \quad (1)$$

If $N \neq \{0\}$, then for all $s, t \in J$,

$$D_H(A_s, A_t)^2 = \|z_s - z_t\|^2 + |\rho_s - \rho_t|^2. \quad (2)$$

If $N = \{0\}$, then $A_t = \{c + z_t\}$ and

$$D_H(A_s, A_t) = \|z_s - z_t\|. \quad (3)$$

Consequently $t \mapsto A_t$ is Hausdorff-continuous on J and locally Lipschitz on I . Quantitatively, for every $0 \leq b < r$ and all s, t such that $\|z_s\|, \|z_t\| \leq b$,

$$D_H(A_s, A_t) \leq \frac{r\|G\|}{\sqrt{r^2 - b^2}} \|s - t\|. \quad (4)$$

On all of J , including its boundary,

$$D_H(A_s, A_t) \leq 2\sqrt{r\|G\| \|s - t\|}. \quad (5)$$

Proof. Write $x - c = m + n$ with $m \in M$, $n \in N$. The condition $Lx = t$ is equivalent to

$$Lm = t - Lc,$$

so it forces $m = z_t$. Orthogonality and Pythagoras give

$$\|x - c\|^2 = \|z_t\|^2 + \|n\|^2.$$

This proves (1), as well as the descriptions of I and J .

Assume first that $N \neq \{0\}$. If $a, b \in M$, $\alpha, \beta \geq 0$, and $n \in N$, orthogonality gives

$$\text{dist}(a + n, b + \beta \mathbb{B}_N)^2 = \|a - b\|^2 + (\|n\| - \beta)_+^2. \quad (6)$$

Taking the supremum over $\|n\| \leq \alpha$ shows

$$D_{\text{asy}H}(a + \alpha \mathbb{B}_N, b + \beta \mathbb{B}_N)^2 = \|a - b\|^2 + (\alpha - \beta)_+^2.$$

Interchanging the sets and taking the maximum proves (2). When $N = \{0\}$, formula (3) is immediate.

Let $d = \|z_s - z_t\|$. If $\|z_s\|, \|z_t\| \leq b < r$, then

$$\begin{aligned} |\rho_s - \rho_t| &= \frac{||z_t\|^2 - \|z_s\|^2|}{\rho_s + \rho_t} \\ &\leq \frac{b}{\sqrt{r^2 - b^2}} d. \end{aligned}$$

Combining this with (2), or just using (3), yields

$$D_H(A_s, A_t) \leq \sqrt{1 + \frac{b^2}{r^2 - b^2}} d = \frac{r}{\sqrt{r^2 - b^2}} d \leq \frac{r\|G\|}{\sqrt{r^2 - b^2}} \|s - t\|,$$

which is (4).

For the boundary estimate, the elementary inequality $(\sqrt{u} - \sqrt{v})^2 \leq |u - v|$ gives

$$|\rho_s - \rho_t|^2 \leq ||z_s\|^2 - \|z_t\|^2| \leq 2rd.$$

Also $d \leq 2r$, hence $d^2 \leq 2rd$. Therefore

$$D_H(A_s, A_t)^2 \leq d^2 + 2rd \leq 4rd \leq 4r\|G\| \|s - t\|,$$

with the singleton case even easier. This proves (5) and all continuity assertions. $\square$

4 Optimal-value stability without ambient compactness

For $f : C \rightarrow \mathbb{R}$, write

$$\phi_-(t) := \inf_{x \in A_t} f(x), \quad \phi_+(t) := \sup_{x \in A_t} f(x).$$

Theorem 2 (Infinite-dimensional optimal-value extension). *In the setting of Theorem 1:*

1. *If $f : C \rightarrow \mathbb{R}$ is uniformly continuous, then f is bounded on C , and both ϕ_- and ϕ_+ are finite and uniformly continuous on J . If f is Λ -Lipschitz, then*

$$|\phi_\pm(s) - \phi_\pm(t)| \leq \Lambda D_H(A_s, A_t), \quad (7)$$

so (4) and (5) give explicit moduli.

2. *If $N = \ker L$ is finite-dimensional and $f : C \rightarrow \mathbb{R}$ is merely continuous, then ϕ_- and ϕ_+ are finite, attained, and continuous on J .*

Thus both compactness of C and finite dimensionality of the ambient space can be removed. In particular, part 1 allows an arbitrary, possibly infinite-dimensional kernel, while part 2 retains the source's mere continuity hypothesis whenever the kernel is finite-dimensional.

Proof. First suppose that f is uniformly continuous. It is bounded on C : choose $\delta > 0$ such that $\|x - y\| < \delta$ implies $|f(x) - f(y)| < 1$, fix $x_0 \in C$, and join x_0 to any $x \in C$ by a line segment split into more than $2r/\delta$ equal pieces. The triangle inequality bounds $|f(x) - f(x_0)|$ uniformly in x .

Now let $s_j \rightarrow s$ in J . Theorem 1 gives $D_H(A_{s_j}, A_s) \rightarrow 0$. Uniform continuity then implies

$$\inf_{A_{s_j}} f \longrightarrow \inf_{A_s} f, \quad \sup_{A_{s_j}} f \longrightarrow \sup_{A_s} f.$$

Indeed, approximate each point of either set by a point of the other and use the uniform modulus; no minimizer is required. Estimate (5) makes this argument uniform on all of J . For a Λ -Lipschitz function it gives the standard estimate (7) directly, with the sharper interior bound (4).

Suppose next that N is finite-dimensional and f is continuous. Each A_t is compact by (1), so its two extrema are finite and attained. Given a sequence $s_j \rightarrow s$, the pairs (z_{s_j}, ρ_{s_j}) converge to (z_s, ρ_s) . The set

$$P := \{(z_s, \rho_s)\} \cup \{(z_{s_j}, \rho_{s_j}) : j \geq 1\}$$

is compact. Therefore

$$K := \bigcup_{j \geq 1} A_{s_j} \cup A_s = \{c + z + \rho n : (z, \rho) \in P, n \in \mathbb{B}_N\}$$

is the continuous image of the compact set $P \times \mathbb{B}_N$, and hence is compact. The restriction $f|_K$ is uniformly continuous, and the first argument applied to $A_{s_j}, A_s \subset K$ proves sequential continuity. Since J is metric, this is continuity. $\square$

5 Sharpness: continuity alone can fail

The finite-dimensional-kernel condition in the second part cannot be removed without some replacement such as uniform continuity.

Proposition 1 (Escaping continuous wells). *Let*

$$H = \mathbb{R} \oplus \ell^2, \quad Y = \mathbb{R}, \quad L(a, x) = a, \quad C = \overline{B}_H(0, 1).$$

Then $I = (-1, 1)$ and

$$A_t = \{(t, x) : \|x\| \leq \sqrt{1 - t^2}\}.$$

There is a bounded continuous $f : C \rightarrow [-1, 0]$ such that

$$\inf_{A_0} f = 0, \quad \inf_{A_{t_n}} f = -1$$

for a sequence $t_n \downarrow 0$. Hence the optimal value function is discontinuous at an interior parameter even though $D_H(A_{t_n}, A_0) \rightarrow 0$. Replacing f by $-f$ gives the same failure for supremal values.

Proof. Let (e_n) be the standard orthonormal basis of ℓ^2 , and put

$$t_n := 2^{-(n+2)}, \quad \rho_n := \sqrt{1 - t_n^2}, \quad y_n := (t_n, \rho_n e_n) \in A_{t_n}, \quad \delta_n := t_n/3.$$

The nearest point of $A_0 = \{0\} \oplus \mathbb{B}_{\ell^2}$ to y_n is $(0, \rho_n e_n)$, so

$$\text{dist}(y_n, A_0) = t_n > \delta_n. \tag{8}$$

For $m \neq n$, orthogonality gives

$$\|y_n - y_m\|^2 = (t_n - t_m)^2 + \rho_n^2 + \rho_m^2.$$

Since $t_n \leq 1/8$, the centers are uniformly separated, whereas $\delta_n \leq 1/24$. Thus the relative balls $B(y_n, \delta_n) \cap C$ are pairwise separated and form a locally finite family.

Define

$$f(x) := \begin{cases} -(1 - \|x - y_n\|/\delta_n), & \text{if } \|x - y_n\| < \delta_n \text{ for some } n, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

The supports are disjoint and locally finite, and the displayed expression vanishes on each support boundary; hence f is continuous on C . Equation (8) implies $f = 0$ on A_0 , while $f(y_n) = -1$. Because $f \geq -1$, the asserted infima follow. Notice also that $x_n = (0, \rho_n e_n) \in A_0$ satisfies $\|x_n - y_n\| = t_n \rightarrow 0$ but $|f(x_n) - f(y_n)| = 1$; this pinpoints the failure of uniform continuity. $\square$

6 Verification, scope, and novelty

The included checker audits the exact Hausdorff formula in finite-dimensional proxies, both quantitative bounds, and the separation and distance claims in Proposition 1. Run

```
conda run --no-capture-output -n sandbox python \
  code/check_hilbert_ball_fibers.py
```

from the packet directory. The computation is only a sanity check; all arguments above are exact.

The source asks for removal “at least in some interesting cases.” The closed-ball class gives a complete affirmative theorem: arbitrary Hilbert dimension, closed-range operators, exact fiber distance, and sharp objective hypotheses. This is not a theorem for all closed bounded convex sets or all Banach spaces. The counterexample rules out the naive statement with an arbitrary continuous objective and an infinite-dimensional kernel, but does not rule out other geometric or coercive hypotheses.

Before promotion, the run indexes were searched for arXiv:1902.02363, the source title, Hilbert balls, affine fibers, Hausdorff continuity, and finite-dimensional kernels. A bounded arXiv/web search used the exact source sentence and close phrases involving affine sections of Hilbert balls and noncompact parametric optimization. It found the source paper and general noncompact Berge-theorem extensions such as [3, 4], but no exact closed-ball fiber formula or sharpness theorem matching Theorems 1–2. This records search bounds; it is not an exhaustive novelty certification.

Human review is recommended. The highest-value checks are the exact asymmetric Hausdorff calculation (6), the compact-union argument for finite-dimensional kernels, and local finiteness in the escaping well construction.

References

- [1] D. Reem, S. Reich, and A. De Pierro, *Stability of the optimal values under small perturbations of the constraint set*, Pure Appl. Funct. Anal. 5 (2020), no. 3, 705–731; arXiv:1902.02363v4. <https://arxiv.org/abs/1902.02363>.
- [2] J. B. Conway, *A Course in Functional Analysis*, second edition, Graduate Texts in Mathematics 96, Springer, 1990.
- [3] E. A. Feinberg, P. O. Kasyanov, and M. Voorneveld, *Berge’s maximum theorem for noncompact image sets*, J. Math. Anal. Appl. 413 (2014), 1040–1046; arXiv:1309.7708.
- [4] E. A. Feinberg, P. O. Kasyanov, and D. N. Kraemer, *Continuity of parametric optima for possibly discontinuous functions and noncompact decision sets*, arXiv:2109.06299, 2021.